

Intermediate Micro - Course Goals

- understand the properties of preferences and choices that let us model consumers as *utility-maximizers*:
 - choose the best (depending on preferences) affordable bundle of goods
- understand how firms make supply decisions, depending on # firms, input costs, anticipated sale price etc
- equilibrium, in-depth
- decision-making under uncertainty
- strategic interaction
- asymmetric information

Preference Relations

- the next few classes will look in detail at how *individual* preferences determine *individual* and *aggregate* demand
- underlying all of this: an assumption that individuals have a well-behaved preference relation, $\succsim$, which determines how they compare any two alternatives:

$a \succsim b$ means “ a is weakly better than b ”

(where a, b represent two possible choices; typically, they'll be bundles of goods)

- other notation of interest:
 - $\succ$ denotes strict preference: we say $a \succ b$ if a is strictly preferred to b (formally, $a \succsim b$ but NOT $b \succsim a$)
 - $\sim$ denotes indifference: we say $a \sim b$ if a and b are equally good (formally, $a \succsim b$ and $b \succsim a$)

Rational Preferences

- in practice, it may be sometimes be difficult to compare two choices, and constraints or confusion may lead to “irrational” preferences and behavior
- but for most of this course, we assume well-behaved, **rational** preferences
- formally, this entails 2 main assumptions on the preference relation:

Assumption 1: Completeness

For any two alternatives a and b , either $a \succsim b$ or $b \succsim a$ or both

- Rephrasing: either you think a is better, or b is better, or you're indifferent
- this rules out someone who is unable to figure out how to compare a and b

Assumption 2: Transitivity

If $a \succsim b$ and $b \succsim c$, then $a \succsim c$

- in other words, preferences must be “logically consistent”, and should not exhibit *cycles* (e.g. where a beats b , b beats c , but suddenly you prefer c to a)
- an example, in words: if Andrea prefers chick flicks to action movies, and action movies to horror movies, then she must prefer chick flicks to horror movies
- otherwise, Andrea might have been tricked out of all her \$ \$ \$ by now: if chick flicks $\succ$ action, action $\succ$ horror, but horror $\succ$ chick flick, you could probably get her to pay you to trade her chick flick in for a horror film, then trade this in for an action film, then trade this back to her original chick flick!

Example

Joe has the following preferences over coffee shops: For any two coffee shops A,B, he weakly prefers A to B if a small cup of coffee is at least as cheap at A as at B, and also A is at least as close (to his house) as B. Thus if p_i, d_i are the price, distance for coffee shop i ,

$$A \succsim B \text{ whenever } p_A \leq p_B \text{ and } d_A \leq d_B$$

Are his preferences rational?

Solution

- we need to check whether preferences are complete and transitive
- transitive? **YES.**
 - take 3 coffee shops A, B, C with $A \succsim B$ and $B \succsim C$: does this imply $A \succsim C$?
 - for Joe, $A \succsim B$ means $p_A \leq p_B$ and $d_A \leq d_B$
 - for Joe, $B \succsim C$ means $p_B \leq p_C$ and $d_B \leq d_C$
 - combining inequalities, $p_A \leq p_C$ and $d_A \leq d_C$
 - so $A \succsim C$, as desired!
- complete? **MAYBE NO!**
 - for example, what if Starbucks is cheaper but Small World is closer?
 - then Joe doesn't think Starbucks $\succsim$ Small world, and also doesn't think Small World $\succsim$ Starbucks

Utility Representations

- our next goal is to see when/whether we can represent preferences $\succsim$, over a set X of options, by a *utility function*
- we're looking for a function $u : X \rightarrow \mathbb{R}$: this means that for each option x in X , the utility function tells us its utility $u(x) \in \mathbb{R}$
- of course, we want the function u to rank alternatives the same way that preferences $\succsim$ do:

Definition

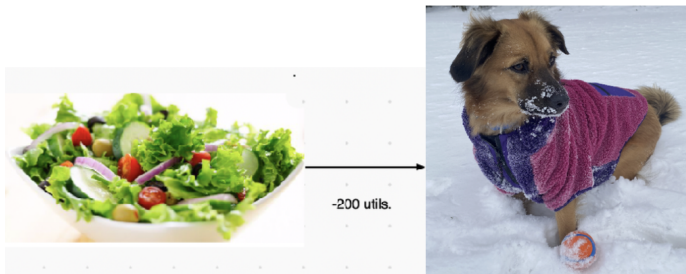
Given a set X and a preference relation $\succsim$, say that u represents $\succsim$ if $\forall x, y \in X$,

$$x \succsim y \Leftrightarrow u(x) \geq u(y)$$

Example



Example Cont'd



More on Interpreting Utility Representations

- we just want the utility function to rank bundles the same way your preferences do
- an equivalent way to write the definition of u represents $\succsim$ (from 3 slides up):
 - $x \succ y \Rightarrow u(x) > u(y)$, and $x \sim y \Rightarrow u(x) = u(y)$
- the actual numbers don't mean much, just the ranking
- for example, suppose $x = (5 \text{ cookies}, 1 \text{ ice cream})$, $y = (3 \text{ cookies}, 2 \text{ ice creams})$, $z = (0 \text{ cookies}, 4 \text{ ice creams})$
- and I tell you that for me, $z \succ y \succ x$
- One representation: $u(z) = 100$, $u(y) = 5$, $u(x) = 1$
- Another representation: $u(z) = 3$, $u(y) = 2$, $u(x) = -100$

Utility Representations: Always Possible?

- assume a rational preference $\succsim$ (otherwise it's for sure impossible)
- if the option set X is finite, obviously yes
 - assign all worst options utility 1, all 2nd worst options utility 2, etc
- less obviously, it is also possible if X is countable
- but if X is infinite uncountable, not always!
- suppose $X = \mathbb{R}^2$ (so a typical option is a pair (x_1, x_2) of real numbers, eg quantities of two goods 1 and 2)
- and consider *lexicographic preferences* on $\mathbb{R}^2$, given by:

$$(x_1, x_2) \succsim (y_1, y_2) \text{ if } x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 \geq y_2$$

Understanding Lexicographic Preferences

- consider two bundles, $x = (x_1, x_2)$ and $y = (y_1, y_2)$
- if one bundle has more of good 1 than the other, you automatically prefer it
- but if the two bundles have the same amount of good 1, then you prefer whichever has more good 2
- for example, you might automatically prefer whichever job pays the most; but for two jobs with equal salaries, pick the one in a better location

Lexicographic Preferences: No Utility Representation

- it turns out, there is no utility function that represents these preferences!
- that is, it is impossible to construct a utility function over bundles of goods, that always ranks them the same way as $\succsim$
- to prove this, suppose, by way of contradiction, that it **is** possible

Lexicographic Proof Cont'd

- for any $x \in \mathbb{R}$, since $(x, 1) \succ (x, 0)$, it must be (if u represents $\succsim$) that $u(x, 1) > u(x, 0)$
- so form the interval $I_x = [u(x, 0), u(x, 1)]$: this has positive width, thus there is room to pick out a rational number from I_x , call it q_x
- now take a bigger number $y > x$, and form the interval $I_y = [u(y, 0), u(y, 1)]$, and pick a rational number q_y from I_y
- since $y > x$, we know $(y, 0) \succ (x, 1)$, so (since u represents $\succsim$) $u(y, 0) > u(x, 1)$
- thus I_x and I_y are disjoint intervals (the left endpoint of I_y is higher than right endpoint of I_x), so $q_y > q_x$
- so we are claiming that any two real numbers can be mapped onto distinct rational numbers
- this is impossible, since the rational numbers are countable while reals are not
- contradiction!

Extra Space for Pictures

The Problem with Lexicographic Preferences

- they are not continuous
- **continuity definition #1:** if $x \succ y$, then there are open neighborhoods $N(x)$ and $N(y)$ such that $x' \succ y'$ for every $x' \in N(x)$ and $y' \in N(y)$
- intuitively, this means that if $x \succ y$, then you can make sufficiently small changes to x and y without reversing the preference
- alternative definition: if $x_n \rightarrow x$ and $y_n \rightarrow y$ and $x_n \succsim y_n \forall n$, then $x \succsim y$
- these definitions fail for lexicographic preferences:
 - eg with 1st definition, $(0, 1) \succ (0, 0)$, but changing $(0, 0)$ to $(\varepsilon, 0)$ reverses the preference, no matter how small ε is
 - eg with 2nd definition, let $x_n = \left(\frac{1}{n}, 0\right)$ and let $y_n = (0, 1)$: then $x_n \succsim y_n \forall n$, but $\lim y_n \succ \lim x_n$

Utility Representation Result

Proposition

(Debreu 1954): If $\succsim$ is a continuous (rational) preference relation on $\mathbb{R}^n$, then there exists a continuous utility function u that represents $\succsim$.

Idea of Proof

- suppose $X = \mathbb{R}$
- now let Z be the set of rational numbers (a countable set that is *dense* in $\mathbb{R}$: this means every interval in $\mathbb{R}$ contains a rational number)
- use continuity to prove that if $x \succ y$, then there is a rational number z_n with $x \succ z_n \succ y$
- now construct utility via:

$$u(x) = \sum_{\{z_n \in Z \mid x \succ z_n\}} \frac{1}{2^n} - \sum_{\{z_n \in Z \mid z_n \succ x\}} \frac{1}{2^n}$$

- easy to verify that $x \succ y \Rightarrow u(x) > u(y)$, and $x \sim y \Rightarrow u(x) = u(y)$
- QED